

Discussion #6 Solutions

Note: Your TA will probably not cover all the problems. This is totally fine; the discussion worksheets are not designed to be finished within an hour. They are deliberately made slightly longer so they can serve as resources you can use to practice, reinforce, and build upon concepts discussed in lectures, labs, and homework.

Relevant Reference Sheet Excerpts

Multiple Linear Regression Formulas

Concept	Formula	Concept	Formula
Mean squared error	$R(\theta) = \frac{1}{n} \ \mathbb{Y} - \mathbb{X}\theta\ _2^2$	Normal equation	$\mathbb{X}^T \mathbb{X} \hat{\theta} = \mathbb{X}^T \mathbb{Y}$
Least squares estimate, if $\mathbb{X}$ is full rank	$\hat{\theta} = (\mathbb{X}^T \mathbb{X})^{-1} \mathbb{X}^T \mathbb{Y}$	Multiple R^2 (coefficient of determination)	$R^2 = \frac{\text{variance of fitted values}}{\text{variance of } y}$
Ridge Regression L2 Regularization	$\frac{1}{n} \ \mathbb{Y} - \mathbb{X}\theta\ _2^2 + \lambda \ \theta\ _2^2$	Squared L2 Norm of $\theta \in \mathbb{R}^d$	$\ \theta\ _2^2 = \sum_{j=1}^d \theta_j^2$
Ridge regression estimate (closed form)	$\hat{\theta}_{\text{ridge}} = (\mathbb{X}^T \mathbb{X} + n\lambda I)^{-1} \mathbb{X}^T \mathbb{Y}$	L1 Norm of $\theta \in \mathbb{R}^d$	$\ \theta\ _1 = \sum_{j=1}^d \theta_j $
LASSO Regression L1 Regularization	$\frac{1}{n} \ \mathbb{Y} - \mathbb{X}\theta\ _2^2 + \lambda \ \theta\ _1$		

Driving with a Constant Model

1. Lillian is trying to use modeling to drive her car autonomously. To do this, she collects `drive` data from driving around her neighborhood. She wants your help to design a model that can drive on her behalf in the future using the outputs of the models you design. First, she wants to tackle two aspects of this autonomous car modeling framework: going forward and turning. Here's the output of `drive.describe()`:

	target_speed	degree_turn
count	531.00000	531.00000
mean	32.923408	143.721153
std	46.678744	153.641504
min	0.231601	0.000000
25%	12.350025	6.916210
50%	25.820689	45.490086
75%	39.788716	323.197168
max	379.919965	359.430309

- Suppose the first part of the model predicts the target speed of the car. Using constant models trained on the speeds of the collected data shown above with L_1 and L_2 loss functions, which of the following is true?
 - ☐ A. The model trained with the L_1 loss will always drive slower than the model trained with L_2 loss.
 - ☐ B. The model trained with the L_2 loss will always drive slower than the model trained with L_1 loss.
 - ☐ C. The model trained with the L_1 loss will sometimes drive slower than the model trained with L_2 loss.
 - ☐ D. The model trained with the L_2 loss will sometimes drive slower than the model trained with L_1 loss.

Solution: When we train a constant model with L_1 loss, the optimal value $\hat{\theta}$ is the median of all the target speeds. From the summary statistics, the median value is indicated by the **50%** percentile value, namely, ≈ 25.82 .

When training with L_2 loss, $\hat{\theta}$ is the mean of all the target speeds, which happens to be ≈ 32.92 . Since the mean is larger than the median, the constant model trained with L_2 loss will always result in a higher speed than when trained with L_1 loss.

- (b) Finding that the model trained with the L_2 loss drives too slowly, Lillian changes the loss function for the constant model where **the loss is penalized more if the true speed is higher**. That way, in order to minimize loss, the model would have to output predictions closer to the true value, particularly as speeds get faster, the end result being a higher constant speed.

Lillian writes this as $L(y, \hat{y}) = y(y - \hat{y})^2$.

Find the optimal $\hat{\theta}_0$ for the constant model using the new empirical risk function $R(\theta_0)$ below:

$$R(\theta_0) = \frac{1}{n} \sum_i y_i (y_i - \theta_0)^2$$

Solution: Take the derivative:

$$\frac{dR}{d\theta_0} = \frac{1}{n} \sum_i \frac{d}{d\theta_0} y_i (y_i - \theta_0)^2 \quad (1)$$

$$= \frac{1}{n} \sum_i -2y_i (y_i - \theta_0) = -\frac{2}{n} \sum_i y_i^2 - y_i \theta_0 \quad (2)$$

Set the derivative to 0:

$$-\frac{2}{n} \sum_i y_i^2 - y_i \theta_0 = 0 \quad (3)$$

$$\theta_0 \sum_i y_i = \sum_i y_i^2 \quad (4)$$

$$\theta_0 = \frac{\sum_i y_i^2}{\sum_i y_i} \quad (5)$$

Note that the empirical risk function is convex, which you can show by computing the second derivative of $R(\theta_0)$ and noting that it is positive for all values of θ_0 .

$$\frac{d^2 R}{d\theta_0^2} = \frac{2}{n} \sum_i y_i > 0$$

since y_i represents speed, also validated by the fact that the minimum value is positive.

Therefore, any critical point must be the global minimum and so the optimal value we found minimizes empirical risk.

- (c) Lillian's friend, Yash, also begins working on a model that predicts the degree of turning at a particular time between 0 and 359 degrees using the data in the `degree_turn` column. Explain why a constant model is likely inappropriate in this use case.

Extra: If you've studied some physics, you may recognize the behavior of our constant model!

Solution: Any constant model will essentially be always turning at an angle and will be unable to turn either direction or go straight (i.e. it'll essentially go in a circle forever).

- (d) Suppose we finally expand our modeling framework to use simple linear regression:

$$f_{\theta}(x) = \theta_{w,0} + \theta_{w,1}x$$

For our first simple linear regression model, we predict the turn angle (y) using target speed (x).

Our optimal parameters are: $\hat{\theta}_{w,1} = 0.019$ and $\hat{\theta}_{w,0} = 143.1$.

However, we realize that we actually want a model that predicts target speed (our new y) using turn angle, our new x (instead of the other way around)!

What are our new optimal parameters $\hat{\theta}_1$ and $\hat{\theta}_0$ for this new model?

Solution: To predict target speed (new y) from turn angle (new x) (what we finally want), we need to compute $\hat{\theta}_1 = \frac{r\sigma_{\text{speed}}}{\sigma_{\text{turn}}}$.

When we predicted the turn angle from target speed (our first SLR model), we computed $\hat{\theta}_{w,1} = \frac{r\sigma_{\text{turn}}}{\sigma_{\text{speed}}}$. To go from $\hat{\theta}_{w,1}$ to $\hat{\theta}_1$, we multiply by $\frac{\sigma_{\text{speed}}^2}{\sigma_{\text{turn}}^2}$.

That is, $\hat{\theta}_1 = \frac{r\sigma_{\text{speed}}}{\sigma_{\text{turn}}} = \left(\frac{r\sigma_{\text{turn}}}{\sigma_{\text{speed}}}\right) \frac{\sigma_{\text{speed}}^2}{\sigma_{\text{turn}}^2} = \hat{\theta}_{w,1} \frac{\sigma_{\text{speed}}^2}{\sigma_{\text{turn}}^2} = 0.019 \cdot \frac{46.678744^2}{153.641504^2} = 0.00175$.

Then, $\hat{\theta}_0 = \bar{y} - \hat{\theta}_1\bar{x} = 32.92 - 0.00175 \cdot 143.72 = 32.67$.

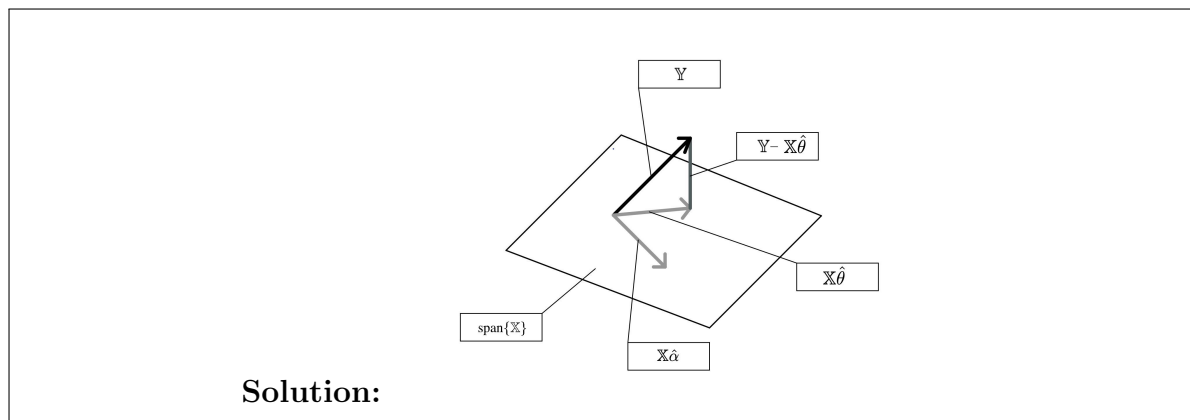
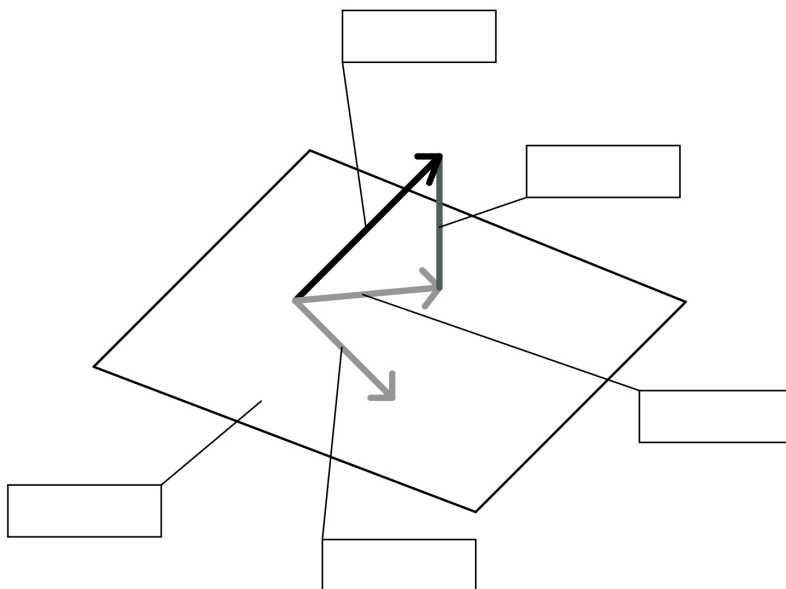
$\bar{y}$ and $\bar{x}$ refer to the means of target speed (our new y) and turn angle (our new x) respectively.

Note: we can't use the inverse function $f_{\theta}^{-1}(x)$ since minimizing the sum of squared verticals is not the inverse problem of minimizing the sum of squared horizontal residuals.

Geometry of Least Squares

2. Suppose we have a dataset represented with the design matrix $\mathbb{X}$ and response vector $\mathbb{Y}$. We use linear regression to solve for this and obtain optimal weights as $\hat{\theta}$. Label the following terms on the geometric interpretation of ordinary least squares:

- $\mathbb{X}$ (i.e., $\text{span}(\mathbb{X})$)
- The response vector $\mathbb{Y}$
- The residual vector $\mathbb{Y} - \mathbb{X}\hat{\theta}$
- The prediction vector $\mathbb{X}\hat{\theta}$ (using optimal parameters)
- A prediction vector $\mathbb{X}\alpha$ (using an arbitrary vector α).



Ordinary Least Squares Potpourri

3. Using the geometry of least squares, let's answer a few questions about Ordinary Least Squares (OLS)!

- (a) Which of the following are true about the optimal solution $\hat{\theta}$ to OLS? Recall that the least squares estimate $\hat{\theta}$ solves the normal equation $(\mathbb{X}^T \mathbb{X})\theta = \mathbb{X}^T \mathbb{Y}$.

$$\hat{\theta} = (\mathbb{X}^T \mathbb{X})^{-1} \mathbb{X}^T \mathbb{Y}$$

- ☐ A. Using the normal equation, we can derive an optimal solution for simple linear regression with an L_2 loss.
- ☐ B. Using the normal equation, we can derive an optimal solution for simple linear regression with an L_1 loss.
- ☐ C. Using the normal equation, we can derive an optimal solution for a constant model with an L_2 loss.
- ☐ D. Using the normal equation, we can derive an optimal solution for a constant model with an L_1 loss.
- ☐ E. Using the normal equation, we can derive an optimal solution for the model $\hat{y} = \theta_1 x + \theta_2 \sin(x^2)$.
- ☐ F. Using the normal equation, we can derive an optimal solution for the model $\hat{y} = \theta_1 \theta_2 + \theta_2 x^2$.

Solution: We can derive solutions to both simple linear regression and constant model with an L_2 loss since they can be represented in the form $y = x^T \theta$ in some way. Specifically, one of the two entries of x would be 1 for SLR (and the other would be the explanatory variable). The only entry for the constant model would be 1.

We cannot derive solutions for anything with the L_1 loss since the normal equation optimizes for MSE.

Since option E is linear with respect to θ , we can use the normal equation. A good rule of thumb to determine if an equation is linear to θ is checking if an expression can separate into a matrix product of two terms:

$$\hat{y} = \begin{bmatrix} x & \sin(x^2) \end{bmatrix} \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}$$

Conversely, option F is not linear in theta and so we cannot use the normal equations.

- (b) Which of the following conditions are required for the least squares estimate in the previous subpart?

- ☐ A. $\mathbb{X}$ must be full column rank.
☐ B. $\mathbb{Y}$ must be full column rank.
☐ C. $\mathbb{X}$ must be invertible.
☐ D. $\mathbb{X}^T$ must be invertible.

Solution: $\mathbb{X}$ must be full column rank in order for the normal equation to have a unique solution. If $\mathbb{X}$ is not full column rank, then $\mathbb{X}^T\mathbb{X}$ will not be invertible, and there will be infinite least squares estimates.

Note that invertibility is required of $\mathbb{X}^T\mathbb{X}$: neither $\mathbb{X}$ nor $\mathbb{X}^T$ need to be invertible (i.e. a counterexample is $\mathbb{X}^T = [1, 0]$).

$\mathbb{Y}$ does not need to be full column rank for a unique least squares estimate.

- (c) What is always true about the residuals in the least squares regression? Select all that apply.

- ☐ A. They are orthogonal to the column space of the design matrix.
☐ B. They represent the errors of the predictions.
☐ C. Their sum is equal to the mean squared error.
☐ D. Their sum is equal to zero.
☐ E. None of the above.

Solution:

(C): (C) is wrong because the mean squared error is the *mean* of the sum of the *squares* of the residuals.

(D): In general, the sum of residuals is not equal to 0 unless the design matrix

$\mathbb{X}$ contains a bias column. A counter-example is: $\mathbb{X} = \begin{bmatrix} 2 & 3 \\ 1 & 5 \\ 2 & 4 \end{bmatrix}$, $\mathbb{Y} = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}$. After

solving the least squares problem, the sum of the residuals is -0.0247 , which is not equal to zero.

(E): is wrong since (A) and (B) are correct.

- (d) Which of the following are true about the predictions made by OLS? Select all that apply.

- ☐ A. They are projections of the observations onto the column space of the design matrix.
☐ B. They are linear combinations of the features.
☐ C. They are orthogonal to the residuals.
☐ D. They are orthogonal to the column space of the features.

☐ E. None of the above.

Solution:

(A) is correct because they are linear projections onto the column space. This fact also makes (C) correct, (E) incorrect, and (D) incorrect.

(B) is correct based on the definition of OLS.

(e) Which of the following is true of the mystery quantity $\vec{v} = (I - \mathbb{X}(\mathbb{X}^T\mathbb{X})^{-1}\mathbb{X}^T)\mathbb{Y}$?

☐ A. The vector $\vec{v}$ represents the residuals for any linear model.

☐ B. If the $\mathbb{X}$ matrix contains the $\vec{1}$ vector, then the sum of the elements in vector $\vec{v}$ is 0 (i.e. $\sum_i v_i = 0$).

☐ C. All the column vectors x_i of $\mathbb{X}$ are orthogonal to $\vec{v}$.

☐ D. If $\mathbb{X}$ is of shape n by p , there are p elements in vector $\vec{v}$.

☐ E. For any $\vec{\alpha}$, $\mathbb{X}\vec{\alpha}$ is orthogonal to $\vec{v}$.

Solution:

(A) is incorrect because any linear model does not create the residual vector v ; only the optimal linear model does.

(D) is incorrect because the vector v is of size n since there are n data points.

The rest are correct by properties of orthogonality as given by the geometry of least squares.